WORK PACKAGE

WORK PACKAGE #	DATE	CLASSIFICATION
30009-10013-705	**12 April 2022**	**RESTRICTED**

TITLE/SUBJECT:

MIL-STD-1553 Dual redundant bus Remote Terminal VHDL core

DISTRIBUTION:

Master: EBD SAP, -----

CONCLUSIONS/DECISIONS/AMENDMENTS:

This work package describes the FPGA firmware implementation for the MIL-STD-1553 dual redundant bus remote terminal (RT) core implementation.

The core is designed and optimized for the Intel (Altera) range of products, however, with simple changes alternative architectures to the memory blocks and PLL components, the implementation can be ported to other vendors.

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PREVIOUS WP #:	**N/A**	SUPERSEDING WP #:	**N/A**
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ABBREVIATIONS

ABBREVIATION	DESCRIPTION
FPGA	Field Programmable Gate Array
ADC	Analog to Digital Converter

1. INTRODUCTION

MIL-STD-1553 is a standard defining a redundant bus topology for use in military equipment. MIL-STD-1553 forms the basis of the communication bus for the MIL-STD-1760 specification.

This core implementation is focussed specifically on the implementation of a dual redundant remote terminal device, implemented on Intel architecture devices.

The top-level architecture of the implementation is shown in Figure 1. The design makes use of two identical remote terminal media access controllers (MAC) and a common CPU and RAM interface. The MAC of each bus connects to the physical layer device (PHY) located on the PCB external to the FPGA.

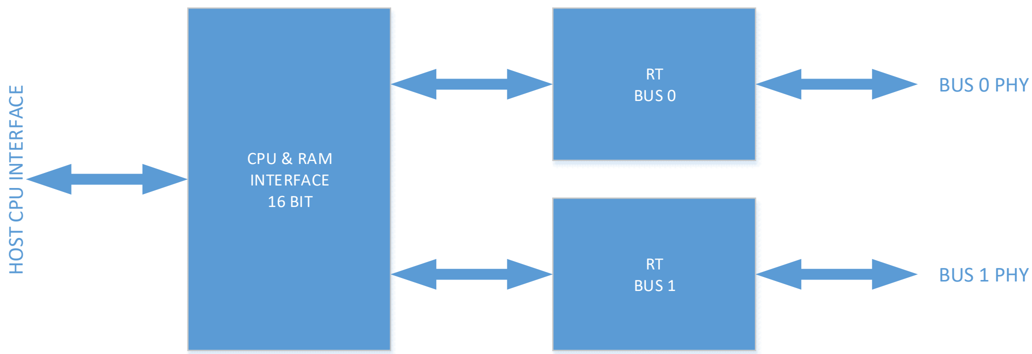

Figure 1: Top level architecture

2. REFERENCE DOCUMENTS

2.1 APPLICABLE DOCUMENTS

[1] MIL-STD-1553C

[2] MIL-HDBK-1553A

[3] MIL-STD-1760E

[4] MIL-HDBK-1760A

3. CONCEPT OF OPERATIONS

The MIL-STD-1553 Standard (hereafter referred to as “Standard”) requires a remote terminal to respond within 150ms of power on to communications from the bus controller. The initial reply may utilize the BusBusy bit to indicate that the RT is not yet ready, however, within 500ms the RT needs to clear the BusBusy bit and reply with the correct message replies as defined in the Standard. Typically, in the application of MIL-STD-1760, the RT shall reply with the correct store identification message. The MAC design is such that it would immediately after FPGA configuration completion be able to start communicating with the host. The maximum internal configuration time for the MAX10 10M16 device using an encrypted compressed image is 22.3ms. This allows the design to meet the 150ms requirement. In order for the design to meet the 500ms requirement, the host processor software needs to define the reply messages by configuring the Transmit RAM location for each message and clear the

BusBusy bit in the RT control register (reg_rt_control). During this initial power on configuration, the host may also configure additional settings for the RT core such as enabling interrupts for received message and error conditions.

The CPU and RAM interface block contains the registers to control the core as well as transmit and receive RAM.

4. MEMORY MAP

4.1 HIGH LEVEL MEMORY MAP

The memory map as indicated in Table 1 shows the main sections of the memory as mapped via the CPU interface. These address areas and values need to be added to the CPU memory offset defined during the instantiation of the core in the host CPU system.

The details of the first four sections for the “Registers” are given in section 4.2 and the transmit and receive RAM in section 0 and in section 4.4 respectively.

Start	End	Size(words)	Function
0x0000	0x000F	16	Common registers
0x0010	0x001F	16	Bus1 Registers
0x0020	0x002F	16	Bus2 Registers
0x0030	0x003F	16	Common registers
0x0040	0x03FF	960	Reserved
0x0400	0x07FF	1024	Transmit RAM
0x0800	0x0BFF	1024	Receive RAM

Table 1: High Level Memory Map

4.2 REGISTERS

The remote terminal core is configured and controlled via the host CPU using a memory mapped interface. The blue coloured registers are general for both busses, the green is bus 1 specific register and the orange are bus2 specific register.

This section describes the CPU interface and registers.

Registers	Address	Read/Write	Default
reg_status	0x0000	Read	0x0000
reg_intr_mask	0x0001	Write	0x0000
reg_clear_bits	0x0002	Read/Write	0x0000
reg_tmr_1us	0x0003	Read/Write	0x0063
reg_tmr_GAP	0x0004	Read/Write	0x018F
reg_tmr_NRP	0x0005	Read/Write	0x0577
reg_tmr_REP	0x0006	Read/Write	0x018F
reg_tmr_SAF	0x0007	Read/Write	0x031F
reg_legal_subaddr1	0x0008	Read/Write	0xFFFF
reg_legal_subaddr2	0x0009	Read/Write	0xFFFF

Registers	Address	Read/Write	Default
reg_legal_rx_mode1	0x000A	Read/Write	0x0000
reg_legal_rx_mode2	0x000B	Read/Write	0x0032
reg_legal_tx_mode1	0x000C	Read/Write	0x01FF
reg_legal_tx_mode2	0x000D	Read/Write	0x000D
reg_legal_brdcst_mode1	0x000E	Read/Write	0x01FA
reg_legal_brdcst_mode2	0x000F	Read/Write	0x0032
reg_status_rxed1	0x0010	Read	0x0000
reg_mode_cmd_rxed1	0x0011	Read	0x0000
reg_mode_data_rxed1	0x0012	Read	0x0000
reg_rx_cmd	0x0013	Read	0x0000
reg_tx_control1	0x0014	Read/Write	0x0000
reg_cmd_proc_start1	0x0015	Read/Write	0x0000
reg_cmd_proc_length1	0x0016	Read/Write	0x0000
reg_cmdrxed1	0x0017	Read	0x0000
reg_24	0x0018		0x0000
reg_25	0x0019		0x0000
reg_26	0x001A		0x0000
reg_27	0x001B		0x0000
reg_28	0x001C		0x0000
reg_29	0x001D		0x0000
reg_bus1_status	0x001E		0x0000
reg_bus1_mask	0x001F		0x0000
reg_status_rxed2	0x0020	Read	0x0000
reg_mode_cmd_rxed2	0x0021	Read	0x0000
reg_mode_data_rxed2	0x0022	Read	0x0000
reg_rx_cmd2	0x0023	Read	0x0000
reg_tx_control2	0x0024	Read/Write	0x0000
reg_cmd_proc_start2	0x0025	Read/Write	0x0000
reg_cmd_proc_length2	0x0026	Read/Write	0x0000
reg_cmdrxed2	0x0027	Read	0x0000
reg_40	0x0028		0x0000
reg_41	0x0029		0x0000
reg_42	0x002A		0x0000
reg_43	0x002B		0x0000
reg_44	0x002C		0x0000
reg_45	0x002D		0x0000
reg_bus2_status	0x002E		0x0000
reg_bus2_mask	0x002F		0x0000
Timestamp3 MSB	0x0030	Read/Write	0x0000
Timestamp2	0x0031	Read/Write	0x0000
Timestamp1	0x0032	Read/Write	0x0000
Timestamp0 LSB	0x0033	Read/Write	0x0000
reg_rt_control	0x0034	Read/Write	0x8000
reg_err_inj_data	0x0035	Read/Write	0x0000

Registers	Address	Read/Write	Default
reg_gID	0x0036	Read/Write	0x0000
reg_fw_version	0x0037	Read	0x0000
reg_rt_addr	0x0038	Read	0x0001
reg_repeat_rate	0x0039	Read/Write	0x0000

Table 2: Registers

4.2.1 reg_status (0x0000)

Read/Write	Default
Read	0x0000

15	14	13	12	11	10	9	8
TxAvail1	TxAvail2	Tx1_Off	Tx2_Off	Inhibit T/F	DBC accepted	Synchronize	Initiate BIT

7	6	5	4	3	2	1	0
Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved

TxAvail1	1 indicates that transmitter 1 is available, 0 indicates that the transmitter 1 is busy.
TxAvail2	1 indicates that transmitter 2 is available, 0 indicates that the transmitter 2 is busy.
Tx1_Off	1 when a transmitter shutdown mode command is received for transmitter 1. This mode command would be received on bus 2. The bit will only be cleared when the clear transmitter shutdown mode command is received on bus 2.
Tx2_Off	1 when a transmitter shutdown mode command is received for transmitter 2. This mode command would be received on bus 1. The bit will only be cleared when the clear transmitter shutdown mode command is received on bus 1.
Inhibit T/F	1 indicates the “Inhibit T/F” mode command has be received. The bit will be cleared if the clear inhibit T/F mode command is received. The bit is a logical OR of the inhibit T/F flag of both MACs (bus 1 and bus 2). Thus both need to be cleared for the bit to be 0.
DBC_accepted	1 indicates that the RT MAC accepted command to become bus controller. For this function to be active, the DBC Enable bit in the reg_rt_control also needs to be set to 1. The bit is a logical OR of both the MACs DBC Accepted bit. (NOTE: the bus controller functionality is not included in this core)
Synchronize	1 indicates that a Synchronize mode command has been received on either bus 1 or bus 2. The bit is latched until cleared by the host. The host can clear this bit by writing a 1 to bit position 11 of the reg_clear_bits register.
InitiateBIT	1 indicates that an Initiate Build in Selftest mode command has been received on either bus 1 or bus 2. The bit is latched until cleared by the host. The host can clear this bit by writing a 1 to bit position 11 of the reg_clear_bits register.
Reserved	Reserved for future use. Value is unspecified and may read back either 0 or 1

A bit that is set in the status register can generate an interrupt to the host CPU if its corresponding bit in the interrupt mask register (reg_intr_mask) is set. The user should ensure that bit 7 down to 0 are cleared in the mask register to avoid spurious interrupts from the lower byte of the status register

4.2.2 reg_intr_mask (0x0001)

Read/Write	Default
Read/Write	0x0000

15	14	13	12	11	10	9	8
TxAvail1	TxAvail2	Tx1_Off	Tx2_Off	Inhibit T/F	DBC accepted	Synchronize	Initiate BIT

7	6	5	4	3	2	1	0
Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved

TxAvail1	1 enables the interrupt to indicate that transmitter 1 is available, 0 disables the interrupt
TxAvail2	1 enables the interrupt to indicate that transmitter 2 is available, 0 disables the interrupt
Tx1_Off	1 enables the interrupt to indicate that transmitter 1 shutdown mode command has been received, 0 disables the interrupt
Tx2_Off	1 enables the interrupt to indicate that transmitter 2 shutdown mode command has been received 0 disables the interrupt
Inhibit T/F	1 enables the interrupt to indicate that either bus 1 or bus 2 received an inhibit T/F mode command, 0 disables the interrupt
DBC_accepted	1 enables the interrupt to indicate that DBC mode command has been accepted by either bus 1 or bus2, 0 disables the interrupt
Synchronize	1 enables the interrupt to indicate that a synchronize mode command was received on either but 1 or bus 2, 0 disables the interrupt
InitiateBIT	1 enables the interrupt to indicate that an initiate BIT mode command has been received on either bus 1 or bus 2, 0 disables the interrupt
Reserved	Reserved for future use. Set to 0 to avoid spurious interrupts.

4.2.3 reg_clear_bits (0x0002)

Read/Write	Default
Read/Write	0x0000

15	14	13	12	11	10	9	8
Reserved	Reserved	Reserved	Reserved	**Synchronize**	**InitiateBIT**	Reserved	Reserved

7	6	5	4	3	2	1	0
Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved

Synchronize	1 commands the MAC to clear the latched bit. The command is sent to both MACs and the results is that the Synchronize bit in the status register (reg_status.9) is cleared. This bit clears automatically and will be read back as 0.
Initiate BIT	1 commands the MAC to clear the latched bit. The command is sent to both MACs and the results is that the InitiateBIT bit in the status register (reg_status.8) is cleared. This bit clears automatically and will be read back as 0.
Reserved	Reserved for future use. Set to 0 to avoid spurious interrupts.

4.2.4 reg_tmr_1us (0x0003)

Read/Write	Default
Read/Write	0x0063

This register controls the timeout of the 1us timer. This 1us timer is used to enable the safety timer (SAF). The value in SAF timer is thus given in increments of the 1us timer value. Typically this timer value should not be modified, but during simulations it is convenient to speed up some timer actions for testing purposes.

The **default (0x0063)** is the correct value for operating from a 100MHz internal clock. The value to set is calculated by:

$$reg\_tmr\_1us = (1us * 100Mhz) - 1$$

4.2.5 reg_tmr_GAP (0x0004)

Read/Write	Default
Read/Write	0x018F

This register controls the intermessage gap time. The specification calls for a minimum intermessage gap of 4us, but also calls for a response of 12us. The GAP timer must thus be set to a value between these limits. Typically this timer value should not be modified, but during simulations it is convenient to speed up some timer actions for testing purposes.

The **default (0x018F)** is the correct value for operating from a 100MHz internal clock. The value to set is calculated by:

$$reg\_tmr\_GAP = (4us * 100Mhz) - 1$$

4.2.6 reg_tmr_NRP (0x0005)

Read/Write	Default
Read/Write	0x0577

This register controls the no response time. The specification calls for a no response time of 14us. Typically this timer value should not be modified, but during simulations it is convenient to speed up some timer actions for testing purposes.

The **default (0x0577)** is the correct value for operating from a 100MHz internal clock. The value to set is calculated by:

$$reg\_tmr\_NRP = (14us * 100Mhz) - 1$$

4.2.7 reg_tmr_REP (0x0006)

Read/Write	Default
Read/Write	0x0195

This register controls the response time to a valid command. The specification calls for a response time between 4us and 12us. The REP timer must thus be set to a value between these limits. Typically this timer value should not be modified, but during simulations it is convenient to speed up some timer actions for testing purposes.

The **default (0x0195)** is the correct value for operating from a 100MHz internal clock. The value to set is calculated by:

$$reg\_tmr\_REP = (4us * 100Mhz) - 1$$

4.2.8 reg_tmr_SAF (0x0007)

Read/Write	Default
Read/Write	0x031F

This register controls the safety timer. The specification calls for a terminal fail safe of 800us. Typically this timer value should not be modified, but during simulations it is convenient to speed up some timer actions for testing purposes.

The **default (0x031F)** is the correct value for operating from a 100MHz internal clock when the reg_tmr_1us is set to the correct value for 1us. The SAF timer increments in 1us intervals. The value to set is calculated by:

$$reg\_tmr\_SAF = \left(\frac{800us}{1us}\right) - 1$$

4.2.9 reg_legal_subaddr1 (0x0008)

Read/Write	Default
Read/Write	0xFFFF

The reg_legal_subaddr1 and reg_legal_subaddr2 are combined to form a 32bit LegalSubAddr value with the reg_legal_subaddr1 being the least significant bits.

Each bit in this LegalSubAddr relates to the sub address sent by the bus controller. The remote terminal will only respond to sub addresses where the bit in this LegalSubAddr register is set to 1.

The **default value (0xFFFF)** allows sub addresses 0 to 15, however bit 0 is ignored since it has the special function of indicating mode commands.

4.2.10 reg_legal_subaddr2 (0x0009)

Read/Write	Default
Read/Write	0xFFFF

The reg_legal_subaddr1 and reg_legal_subaddr2 are combined to form a 32bit LegalSubAddr value with the reg_legal_subaddr1 being the least significant bits.

Each bit in this LegalSubAddr relates to the sub address sent by the bus controller. The remote terminal will only respond to sub addresses where the bit in this LegalSubAddr register is set to 1.

The **default value (0xFFFF)** allows sub addresses 16 to 32, however bit 31 is ignored since it has the special function of indicating mode commands.

4.2.11 reg_legal_rx_mode1 (0x000A)

Read/Write	Default
Read/Write	0x0000

The reg_legal_rx_mode1 and reg_legal_rx_mode2 are combined to form a 32bit LegalModeRx value with the reg_legal_rx_mode1 being the least significant bits.

Each bit in this LegalModeRx relates to the mode command sent by the bus controller where the controller expects a data word reply. The remote terminal will only respond to mode commands where the bit in this LegalModeRx register is set to 1.

The **default value (0x0000)** is as per TABLE I in [1].

4.2.12 reg_legal_rx_mode2 (0x000B)

Read/Write	Default

Read/Write	0x0032

The reg_legal_rx_mode1 and reg_legal_rx_mode2 are combined to form a 32bit LegalModeRx value with the reg_legal_tx_mode1 being the least significant bits.

Each bit in this LegalModeRx relates to the mode command sent by the bus controller where the controller expects a data word reply. The remote terminal will only respond to mode commands where the bit in this LegalModeRx register is set to 1.

The **default value (0x0032)** is as per TABLE I in [1], allowing the following mode commands:

- Synchronize
- Selected Transmitter shutdown
- Override Selected Transmitter shutdown

4.2.13 reg_legal_tx_mode1 (0x000C)

Read/Write	Default
Read/Write	0x01FF

The reg_legal_tx_mode1 and reg_legal_tx_mode2 are combined to form a 32bit LegalModeTx value with the reg_legal_rx_mode1 being the least significant bits.

Each bit in this LegalModeTx relates to the mode command sent by the bus controller. The remote terminal will only respond to mode commands where the bit in this LegalModeTx register is set to 1.

The **default value (0x01FF)** is as per TABLE I in [1], allowing the following mode commands:

- Dynamic Bus control
- Synchronize
- Transmit Status Word
- Initiate Self Test
- Transmitter Shutdown
- Override Transmitter Shutdown
- Inhibit Terminal Flag Bit
- Override Inhibit Terminal Flag Bit
- Reset Remote Terminal

4.2.14 reg_legal_tx_mode2 (0x000D)

Read/Write	Default
Read/Write	0x000D

The reg_legal_tx_mode1 and reg_legal_tx_mode2 are combined to form a 32bit LegalModeTx value with the reg_legal_rx_mode1 being the least significant bits.

Each bit in this LegalModeTx relates to the mode command sent by the bus controller. The remote terminal will only respond to mode commands where the bit in this LegalModeTx register is set to 1.

The **default value (0x000D)** is as per TABLE I in [1], allowing the following mode commands:

- Transmit Vector Word
- Transmit Last Command
- Transmit BIT word

4.2.15 reg_legal_brdcst_mode1 (0x000E)

Read/Write	Default
Read/Write	0x01FA

The reg_legal_brdcst_mode1 and reg_legal_brdcst_mode2 are combined to form a 32bit LegalModeBrdcst value with the reg_legal_brdcst_mode1 being the least significant bits.

Each bit in this LegalModeBrdcst relates to the mode command sent by the bus controller. The remote terminal will only respond to mode commands where the bit in this LegalModeBrdcst register is set to 1 when the station ID is the broadcast station ID.

The **default value (0x01FA)** is as per TABLE I in [1], allowing the following broadcast mode commands:

- Synchronize
- Initiate Self Test
- Transmitter Shutdown
- Override Transmitter Shutdown
- Inhibit Terminal Flag Bit
- Override Inhibit Terminal Flag Bit
- Reset Remote Terminal

4.2.16 reg_legal_brdcst_mode2 (0x000F)

Read/Write	Default
Read/Write	0x0032

The reg_legal_brdcst_mode1 and reg_legal_brdcst_mode2 are combined to form a 32bit LegalModeBrdcst value with the reg_legal_brdcst_mode1 being the least significant bits.

Each bit in this LegalModeBrdcst relates to the mode command sent by the bus controller. The remote terminal will only respond to mode commands where the bit in this LegalModeBrdcst register is set to 1 when the station ID is the broadcast station ID.

The **default value (0x0032)** is as per TABLE I in [1], allowing the following broadcast mode commands:

- Synchronize
- Selected Transmitter shutdown
- Override Selected Transmitter shutdown

4.2.17 reg_status_rxed1 (0x0010) and reg_status_rxed2 (0x0020)

These two registers are identical in function and each relates to the status word received on its bus. The ParityOK bit (bit 7) is replaced with the currently calculated parity of the address wires.

Read/Write	Default
Read	N/A

15	14	13	12	11	10	9	8
RtAddr4	RtAddr3	RTAddr2	RtAddr1	RtAddr0	MessageError	Instrument	ServiceReq

7	6	5	4	3	2	1	0
RtAddrError	Reserved	Reserved	Brdcst Rxed	Busy	SubSysFlag	DBC Acpt	TerminalFlag

RtAddr(4..0)	Contains the address of the transmitting node
Message Error	Indicates that there was an error in the preceding data words.
Instrument	Reserved for instrumentation bit and should always be 0
Service Request	When received by a bus controller, this bit indicates that the RT or associated subsystem requires service
RtAddrError	Indicates an error on local address lines
Brdcst_Rxed	A logic 1 indicates that the preceding valid command word was a broadcast command and a logic 0 shows it was not a broadcast command.
Busy	A logic 1 indicates that the RT is still busy and not ready to respond to new commands.
SubSysFlag	This bit shall flag a subsystem fault condition, and alert the bus controller to potentially invalid data.

DBC_Acpt	A logic one shall indicate acceptance of control, and a logic zero shall indicate rejection of control.
Terminal Flag	This bit shall flag a RT fault condition
Reserved	Reserved for future use.

4.2.18 reg_mode_cmd_rxed1 (0x0011) and reg_mode_cmd_rxed2 (0x0021)

These two registers contain the exact mode command received on each bus.

Read/Write	Default
Read	N/A

4.2.19 reg_mode_data_rxed1 (0x0012) and reg_mode_data_rxed2 (0x0022)

The contents of this register depend on the mode command code. These registers contain the received mode command data word verbatim.

Read/Write	Default
Read	N/A

4.2.20 reg_rx_cmd1 (0x0013) and reg_rx_cmd2 (0x0023)

These registers contain a latched version of the last command received and should be used in conjunction with the NewCmdReceived interrupt.

Read/Write	Default
Read	N/A

15	14	13	12	11	10	9	8
Reserved	Reserved	Reserved	OtherTxOff	Inhibit T/F	DBC	Synchronize	InitiateBIT

7	6	5	4	3	2	1	0
Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved

OtherTxOff	State of other transmitter. If received on Bus 1, indicates the status of the Bus2 transmitter. If 1, turn the transmitter off, if 0 turn the transmitter on.
Inhibit T/F	This command shall cause the RT to control T/F bit. When 1, the T/F bit is inhibited, when 0, it is not inhibited.
DBC	Dynamic Bus control command received
Synchronize	This command shall cause the RT to synchronize
Initiate BIT	This command shall be used to initiate self test within the RT
Reserved	Reserved for future use.

4.2.21 reg_bus1_status (0x001E) and reg_bus2_status (0x002E)

Read/Write	Default
Read	N/A

15	14	13	12	11	10	9	8
Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved

7	6	5	4	3	2	1	0
ModeCmd	DataReq	DataReceived	StatusRxedFlag	SAF	REP	NRP	GAP

ModeCmd	Indicates that a ModeCmd word has been received
DataReq	Indicates that a data word is requested. The host should ensure that the expected data word is updata.
DataReceived	Indicates that the RT received data. The host should read the data from the Rx RAM.
StatusRxedFlag	Indicates that a status word has been received
SAF	Indicates that the SAF safety timer has expired
REP	Indicates that the REP response timer has expired
NRP	Indicates that the NRP No response timer has expired
GAP	Indicates that the GAP intermessage gap timer has expired
Reserved	Reserved for future use

A bit that is set in the status register can generate an interrupt to the host CPU if its corresponding bit in the interrupt mask register (reg_bus#_mask) is set. The user should ensure that bit 15 down to 8 are cleared in the mask register to avoid spurious interrupts from the lower byte of the status register

4.2.22 reg_bus1_mask (0x001F) and reg_bus2_mask (0x002F)

Read/Write	Default
Read/Write	N/A

15	14	13	12	11	10	9	8
Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved

7	6	5	4	3	2	1	0
ModeCmd	DataReq	DataReceived	StatusRxedFlag	SAF	REP	NRP	GAP

ModeCmd	1 enables interrupt generation for received mode command
DataReq	1 enables interrupt generation for data requested mode command
DataReceived	1 enabled interrupt generation for received data
StatusRxedFlag	1 enables interrupt generation for the receipt of a status word
SAF	1 enables interrupt generation for the expiration of the SAF timer

REP	1 enables interrupt generation for the expiration of the REP timer
NRP	1 enables interrupt generation for the expiration of the NRP timer
GAP	1 enables interrupt generation for the expiration of the GAP timer
Reserved	Reserved for future use. Set to 0 to avoid spurious interrupts

4.2.23 Timestamp (0x0030 – 0x0033)

These four registers contain the core 64-bit system time counter. After reset of the core, this counter increments on each clock cycle. When the maximum value is reached, the counter wraps around to zero again.

The address 0x0030 contains the most significant word and the address 0x0033 the least significant word.

4.2.24 reg_rt_control (0x0034)

Read/Write	Default
Read/Write	N/A

15	14	13	12	11	10	9	8
BusBusy	BrdcstEn	DBC_En	Active Bus	Use BC Repeat	BC Repeat Stop	BC Repeat Start	Reserved

7	6	5	4	3	2	1	0
Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved

BusBusy	This bit is inserted into the status word of this RT. The Host needs to clear this bit after completing configuration.
BrdcstEn	This bit control the validity of broadcast mode
DBC_En	This bit will enable the dynamic bus control mode. Not implement.
Use BC Repeat	Enables the BC repeat timer to repeat the current configured command set according to the set interval (reg_repeat_rate)
BC Repeat Start	Start/Reset the BC repeat timer
BC Repeat Stop	Stops the BC repeat timer
Reserved	Reserved for future use. Set to 0 to avoid spurious interrupts

4.2.25 reg_gID (0x0036)

Read/Write	Default
Read/Write	0xBEEF

This register can be used by the host to identify a specific core in case of multiple core instantiations. The host is also free to use this register as a general scratch register. The contents are not used within the core.

4.2.26 reg_fw_version (0x0037)

Read/Write	Default
Read/Write	N/A

This register indicates the current version of the firmware for the core. The MSB nibble indicates the major version and the LSB nibble the minor version.

4.2.27 reg_rt_addr (0x0038)

Read/Write	Default
Read/Write	N/A

This register contains the logical states of the actual RT address lines and parity lines as connected to the core.

15	14	13	12	11	10	9	8
Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved

7	6	5	4	3	2	1	0
Reserved	Reserved	RtAddrParity	RtAddr5	RtAddr3	RtAddr2	RtAddr1	RtAddr0

4.2.28 reg_repeat_rate (0x0039)

Read/Write	Default
Read/Write	N/A

This register contains the repeat rate in increments of 500us.

4.3 TRANSMIT RAM

Start	End	Size	Function
0x0400	0x041F	32	Reserved
0x0420	0x043F	32	SubAddr1
0x0440	0x045F	32	SubAddr2
0x0460	0x047F	32	SubAddr3
0x0480	0x049F	32	SubAddr4
0x04A0	0x04BF	32	SubAddr5
0x04C0	0x04DF	32	SubAddr6
0x04E0	0x04FF	32	SubAddr7
0x0500	0x051F	32	SubAddr8
0x0520	0x053F	32	SubAddr9
0x0540	0x055F	32	SubAddr10
0x0560	0x057F	32	SubAddr11
0x0580	0x059F	32	SubAddr12
0x05A0	0x05BF	32	SubAddr13
0x05C0	0x05DF	32	SubAddr14
0x05E0	0x05FF	32	SubAddr15
0x0600	0x061F	32	SubAddr16
0x0620	0x063F	32	SubAddr17
0x0640	0x065F	32	SubAddr18
0x0660	0x067F	32	SubAddr19
0x0680	0x069F	32	SubAddr20
0x06A0	0x06BF	32	SubAddr21
0x06C0	0x06DF	32	SubAddr22
0x06E0	0x06FF	32	SubAddr23
0x0700	0x071F	32	SubAddr24
0x0720	0x073F	32	SubAddr25
0x0740	0x075F	32	SubAddr26
0x0760	0x077F	32	SubAddr27
0x0780	0x079F	32	SubAddr28
0x07A0	0x07BF	32	SubAddr29
0x07C0	0x07DF	32	SubAddr30
0x07E0	0x07FF	32	MODE Data

Table 3: Transmit RAM Memory Map

The Transmit RAM is divided into 32 blocks of 32 words each. Each block is dedicated to a specific sub-address. The host shall ensure that the data within these blocks are correctly configured. The core will read the data from this location as needed.

Tx RAM address 0x0400 is reserved since sub-address 0 is used for command words.

Tx RAM address 0x07E0 contains the data of the mode commands when the mode command requests a data word.

4.4 RECEIVE RAM

Start	End	Size	Function
0x0800	0x081F	32	Not used
0x0820	0x083F	32	SubAddr1
0x0840	0x085F	32	SubAddr2
0x0860	0x087F	32	SubAddr3
0x0880	0x089F	32	SubAddr4
0x08A0	0x08BF	32	SubAddr5
0x08C0	0x08DF	32	SubAddr6
0x08E0	0x08FF	32	SubAddr7
0x0900	0x091F	32	SubAddr8
0x0920	0x093F	32	SubAddr9
0x0940	0x095F	32	SubAddr10
0x0960	0x097F	32	SubAddr11
0x0980	0x099F	32	SubAddr12
0x09A0	0x09BF	32	SubAddr13
0x09C0	0x09DF	32	SubAddr14
0x09E0	0x09FF	32	SubAddr15
0x0A00	0x0A1F	32	SubAddr16
0x0A20	0x0A3F	32	SubAddr17
0x0A40	0x0A5F	32	SubAddr18
0x0A60	0x0A7F	32	SubAddr19
0x0A80	0x0A9F	32	SubAddr20
0x0AA0	0x0ABF	32	SubAddr21
0x0AC0	0x0ADF	32	SubAddr22
0x0AE0	0x0AFF	32	SubAddr23
0x0B00	0x0B1F	32	SubAddr24
0x0B20	0x0B3F	32	SubAddr25
0x0B40	0x0B5F	32	SubAddr26
0x0B60	0x0B7F	32	SubAddr27
0x0B80	0x0B9F	32	SubAddr28
0x0BA0	0x0BBF	32	SubAddr29
0x0BC0	0x0BDF	32	SubAddr30
0x0BE0	0x0BFF	32	MODE Data

Table 4: Receive RAM Memory Map

The Receive RAM is divided into 32 blocks of 32 words each. Each block is dedicated to a specific sub-address. The host shall read the data from this location when indicated by data received interrupt.

Rx RAM address 0x0800 is reserved since sub-address 0 is used for command words.

Rx RAM address 0x0BE0 contains the data of the mode commands when the mode command contains a data word.

5. REMOTE TERMINAL CONFIGURATION

After power on reset (nReset) signal is released high, the core registers are configured to default values. The core will perform a parity check on the connected RT address lines and parity line. If a parity error is present, the core will remain in reset, but not the CPU interface. The host can check for this condition by investigating reg_rt_addr to see the values of the connected signals as well as reg_status_rxed#.RtAddrError.[1]

The host needs to configure the transmit RAM with correct message data. Once this is complete, the host needs to clear the reg_rt_control.BusBusy bit.

The host also needs to configure the interrupts as required.

5.1 INITIALISATION

The below commands will initialize the RT.

Rd/Wr	Reg Addr	Value	Function
Wr	0x0001	0x0000	Disable all interrupts
Rd	0x0037		Check the firmware Version
Wr	0x0003	0x0063	Configure 1us Timer
Wr	0x0004	0x018F	Configure GAP Timer
Wr	0x0005	0x0577	Configure NRP Timer
Wr	0x0006	0x0195	Configure REP Timer
Wr	0x0007	0x031F	Configure SAF Timer
Wr	0x0008	0xFFFF	Set the legal sub addresses 1
Wr	0x0009	0xFFFF	Set the legal sub addresses 2
Wr	0x000A	0x0000	Set the legal Rx MODE 1
Wr	0x000B	0x0032	Set the legal Rx MODE 2
Wr	0x000C	0x1FF	Set the legal Tx MODE 1
Wr	0x000D	0x000D	Set the legal Tx MODE 2
Wr	0x000E	0x01FA	Set the legal Broadcast MODE 1
Wr	0x000F	0x0032	Set the legal Broadcast MODE 2
Wr	0x001F	0x0000	Clear the interrupt mask for bus 1
Wr	0x002F	0x0000	Clear the interrupt mask for bus 2
Wr	0x0034	0xC000	Configure the general RT control register: - Indicate Bus Busy state - Enable broadcast mode
Wr	0x0015	0x0000	Configure the Bus 1 command processor start address
Wr	0x0016	0x0001	Configure the Bus 1 command processor length
Wr	0x0025	0x0000	Configure the Bus 1 command processor start address
Wr	0x0026	0x0001	Configure the Bus 1 command processor length
Configure the default contents of the different transmit messages for each of the sub addresses and modes used i.e weapon description, fuze etc.			
Wr	0x0034	0x4000	Clear the busy bit.
Wr	0x001F	0x00FF	Set Bus 1 interrupts mask

[1] The # refers to the bus, bus 1 or 2.

			- 0xFF sets all interrupt on, user can disable some of the timer interrupts if needed. - When interrupt is received, writing to register 0x001E clears the interrupt
Wr	0x002F	0x00FF	Set Bus 2 interrupts mask - 0xFF sets all interrupt on, user can disable some of the timer interrupts if needed. - When interrupt is received, writing to register 0x002E clears the interrupt

5.2 BC REQUEST DATA FROM RT

When the RT receives a request for data from the BC, the "DataReq" bit would be set in the status register. If the interrupt mask bit is enables, the core will generate an interrupt to the host. Please note that to determine which bus caused the interrupt, the host can check the general status register at 0x0000.

Rd/Wr	Reg Addr	Value	Function
Wr	0x001F	0x00F2	Set the interrupt mask for bus 1 - ModeCmd enabled - **DataReq enabled** - DataReceived enabled - StatusRxedFlag enabled - NRP (no response timer) enabled
Wr	0x002F	0x00F2	Set the interrupt mask for bus 1 - ModeCmd enabled - **DataReq enabled** - DataReceived enabled - StatusRxedFlag enabled - NRP (no response timer) enabled
Rd	0x0000		Check source of Interrupt: - Bit 0 = Bus 1 - Bit 1 = Bus 2
Rd	0x001E or 0x002E		Check source of Bus1/2 interrupt: - For Data request from BC, bit 6 should be set.
Wr	0x001E or 0x002E	0xFFFF	Clear all set interrupts
The core will automatically reply after timeout of REP with the data in the relevant transit register. Thus, if the host want to update the information, it should be done ASAP before REP expires. To update the data, please refer next steps.			
Rd	0x0017 or 0x0027		Returns the last received command word - Check the sub address and data length, update data if needed.
To update the data, calculate the transmit memory offset to write new data to as follows: 0x0400 plus the Sub address shifted left by 5 bits, thus for sub addr 0x01 the correct address would be 0x0420.			
Wr	0x0420		Write the data as required to the Transmit RAM.

5.3 BC SENDS DATA TO RT

When the BC sends data to the RT, the core will accept the data (if address matches) and save it to the Receive buffers and then set the data received flag in the relevant bus status register. Depending on the configuration of the interrupt mask register, an interrupt will be generated to the host. The host needs to read the data from the buffer. If a new message is received to the same sub address, the data in the receive buffer will be overwritten.

Rd/Wr	Reg Addr	Value	Function
Wr	0x001F	0x00F2	Set the interrupt mask for bus 1 - ModeCmd enabled - DataReq enabled - **DataReceived enabled** - StatusRxedFlag enabled - NRP (no response timer) enabled
Wr	0x002F	0x00F2	Set the interrupt mask for bus 2 - ModeCmd enabled - DataReq enabled - **DataReceived enabled** - StatusRxedFlag enabled - NRP (no response timer) enabled
Rd	0x0000		Check source of Interrupt: - Bit 0 = Bus 1 - Bit 1 = Bus 2
Rd	0x001E or 0x002E		Check source of Bus1/2 interrupt: - For Data request from BC, bit 6 should be set.
Wr	0x001E or 0x002E	0xFFFF	Clear all set interrupts
Rd	0x0017 or 0x0027		Returns the last received command word - Check the sub address and data length, update data if needed.
To read the data, calculate the receive memory offset to read new data from as follows: 0x0800 plus the Sub address shifted left by 5 bits, thus for sub addr 0x01 the correct address would be 0x0820.			
Rd	0x0820		Read the data as required from the Receive RAM.